



FlightGear 2020.3

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F

80-0060 Block 27

Wind tunnel aerodynamics: ref: AFIT/GAE/ENY/90D-16

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V1.11 2019-11-01

Tweaked by Cromha



Cromha's F-15C/EX

Pilot's Manual

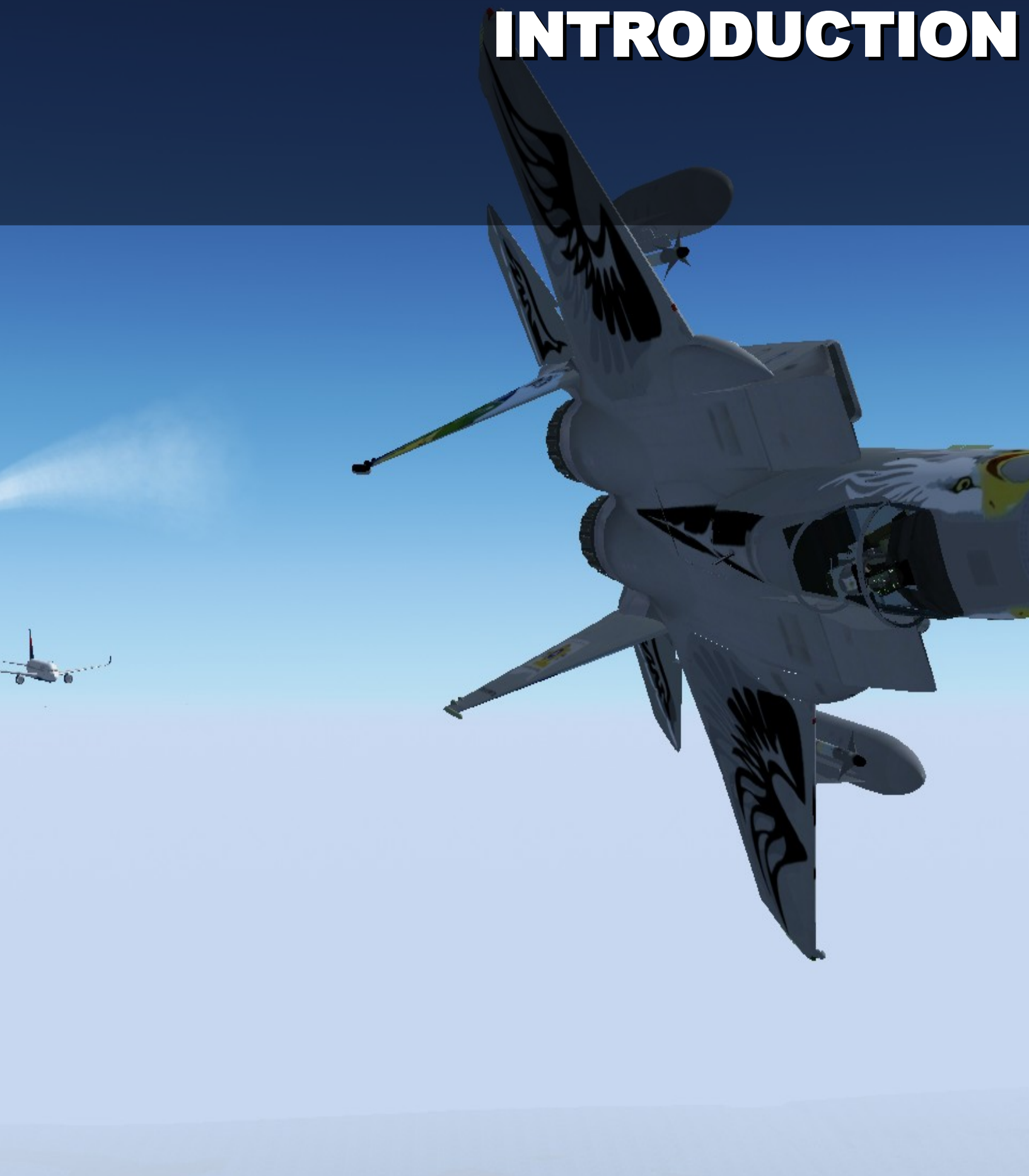
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INTRODUCTION



F-15C INTRODUCTION

The McDonnell Douglas (now Boeing) F-15 Eagle is an American twin-engine, all-weather tactical fighter designed by McDonnell Douglas to gain and maintain air superiority in aerial combat during the Cold War. Following reviews of proposals, the USAF selected McDonnell Douglas's design in 1967 to meet the service's need for a dedicated air superiority fighter. The Eagle first flew in July 1972, and entered service in 1976.

The F-15 Eagle boasts an all-service kill ratio of 140:0 (104 kills for 0 deaths). The F-15 downed the majority of Iraqi aircraft during the Gulf War of 1991, and the same in the Balkan conflict of 1999.

The F-15 was originally envisioned as a pure air superiority aircraft. Its design included a secondary ground-attack capability that was largely unused in the C and D variants. For this reason only air to air missiles are currently supported.

The design proved flexible enough that an all-weather strike derivative, the F-15EX Eagle II, was later developed, entering service in 2024. The F-15 Eagle is expected to be in service with the U.S. Air Force past 2025. The F-15 production line was initially set to end in 2019, 47 years after the type's first flight.

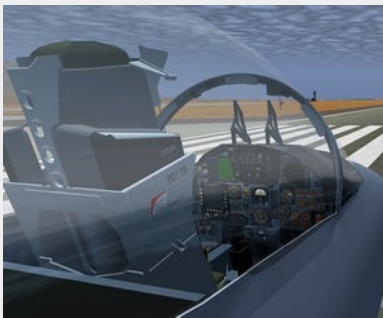
The currently modeled variants of the F-15 in this FlightGear Addon are the F-15C, the F-15D (two-seater variant) and the F-15EX.

Note that all data provided in the manual does not necessarily apply to real life, this is a simulator after all.

FGADDON'S OVERVIEW




External view



Cockpit view

- **Type:** All-weather tactical interceptor fighter
- **Propulsion:** Twin jet-engine
- **Manufacturer:** McDonnell Douglas, Boeing
- **Author(s):** Richard Harrison, Enrique Laso, Gene Buckle, Jimmy L. Miles (Cromha)
- **FDM:** JSBSim
- **--aircraft=** f15c f15d f15db5 f15e
- **Status:** Advanced production
 - **FMD:** 5/5
 - **Systems:** 5/5
 - **Cockpit:** 5/5
 - **Model:** 5/5
 - **Supports:** Dual Control, Aerial Refueling, Rembrandt-compatible, Canvas
- **Development**
 - **Repository:** <https://github.com/OcelotWalrus/F-15>
 - **License:** GPLv2+

Features

- F-15C, D and EX variants
- Realistic FDM based on windtunnel-derived aerodynamic data found in (AFIT/GAE/ENY/90D-16)
- Photorealistic 3D cockpit with real cockpit textures provided by Gene Buckle from his unique F-15C Eagle Flight Simulator Project
- Operational weapons and radar.
- Full range of sound simulation (buffet, gear, engines, etc.). Cockpit aural warnings including voice.
- Canvas radar and HUD, VSD, TEWS, and MPCD.
- Refuelling and stores.
- In-cockpit engine start (including JFS).
- Modelled hydraulic and electrical systems.
- ALS afterburner flames. These flames are Rembrandt-compatible and adjust to take into account ambient lighting.
- External and internal Rembrandt-compatible lighting.
- Modelled ECS, including pressurisation, windscreen temperature, and frosting.
- Control actuator system.
- Overload Warning System.
- Caution/warning system.

Simulation Fidelity and Limitations

- The windtunnel data is valid for a clean configuration at Mach 0.6. The aerodynamic model has been adapted to the entire range of the aircraft, Mach 0 to Mach 2.5.
- Forces and moments for flaps, gear, stores, have been added based on parameter estimation and by using data from other reports. These been validated by comparison with plots from various reports and results are within a tolerance that is acceptable for a desktop simulation.
- ECS, HYDS and Electrics systems models are simplified and do not accurately follow the schematics for the aircraft.
- Timings for gear, cockpit opening are based on observation of video evidence.
- The MPCD is restricted to only implement a few pages, notable PACS
- I have not attempted to accurately model in any way the TEWS system, as all details on it are restricted by security classification.
- Missile and bombs dynamics and tracking simulation is simplistic (based on the work by xiii, and Shinobi), AIM-9 is most accurate, with AIM-7 and AIM-120 being not quite so realistic
- Flight controls and the CAS system are reasonably well modelled; but again accurate data isn't readily available so this is based largely on a published article and knowledge gained from technical reports.

Cromha's Enhancements

- The creation of the EX variant (cockpit model is still old F-15C)
- The enhancement of the C variant cockpit (added simulation of chaff/flare release and low ammo lights, engine flame out oral warning + lights with a system)
- The implementation of A/G unguided operations (CCIP) and racks as well
- Added more payload options, through custom rack count or new stores
- The addition of new A/A missiles: AIM-9M, AIM-9X (only EX variant), and the AIM-120D (only EX variant)
- The addition of new A/G bombs: MK-82, MK-83, MK-84, CBU-87, CBU-105 (only EX variant), and B61-12 (only EX variant)
- The addition of misc weapons: LAU-68C (only EX variant)
- The addition of more aerodynamic effects such as mach cone or strake and G vortex
- The polishing of already-existing effects
- The addition of new real-life and fictional liveries
- The complete overhaul of the in-game F-15 GUI by centralizing controls into the F-15 Eagle Configuration Panel, allow the control of fuel, conformal tanks, different parts' damage, ground services, smoke pods' color, pilot helmet sound proofing, selection of pre-made loads setup or customization of every hardpoint, custom load of gun rounds and flares/chaffs, and useful info on the current F-15 setup
- The polishing of various sound effects (gun, flares etc.)
- Enhanced the HUD to have clearer calibrated airspeed shown, as well as ground speed with it, AOA, clearer current altitude, ft/s indicator below altitude as a landing help, the currently selected armament type if ARM ON, and the chaff and flare count if ARM ON, and display the last launched missiles with a diamond to track their path
- Adding CCIP bombing mode
- Fixed liveries whose nose was bugged
- Tweaked some missiles' data to fit more realistic scenarios (or not)
- The addition of custom FlightGear AI scenarios for either strike, target interception or dogfight situations
- Added a dragchute
- Added 'I' keybind to toggle emergency flare/chaff release (10/sec instead of 2/sec)

FLIGHT CONTROLS

ARI

The ARI is a system that automatically applies rudder based on aileron movement (designed to work as an aid as it's what a good pilot should do anyway). The schedule adjusts based on configuration and airspeed. The system is disabled above Mach 0.9 or whenever the landing gear is deployed.

CAS

The CAS system is a three channel system (Yaw, Pitch, Roll). The YAW channel is basically a yaw damper, the pitch channel is basically a pitch damper and the roll channel is a roll rate limiter. It is not recommended to fly with the YAW CAS channel disabled and is liable to result in serious yaw oscillation at higher mach. If any of these systems are disabled (or inoperative) a warning light will be visible on the caution / warning panel.

Pitch Ratio & Roll Ratio

The CADC adjusts the amount of surface movement of the elevator, ailerons and differential elevator per inch of stick movement based on the airspeed to improve the controllability at higher speeds. Both of these systems can be turned off at the roll ratio is on the MISC panel on the left console, the pitch ratio switch is on the centre panel next to the landing gauge. There is also a gauge on this panel which indicates the current pitch ratio either .5 in the emergency mode or the value determined by the CADC.

AURAL WARNINGS

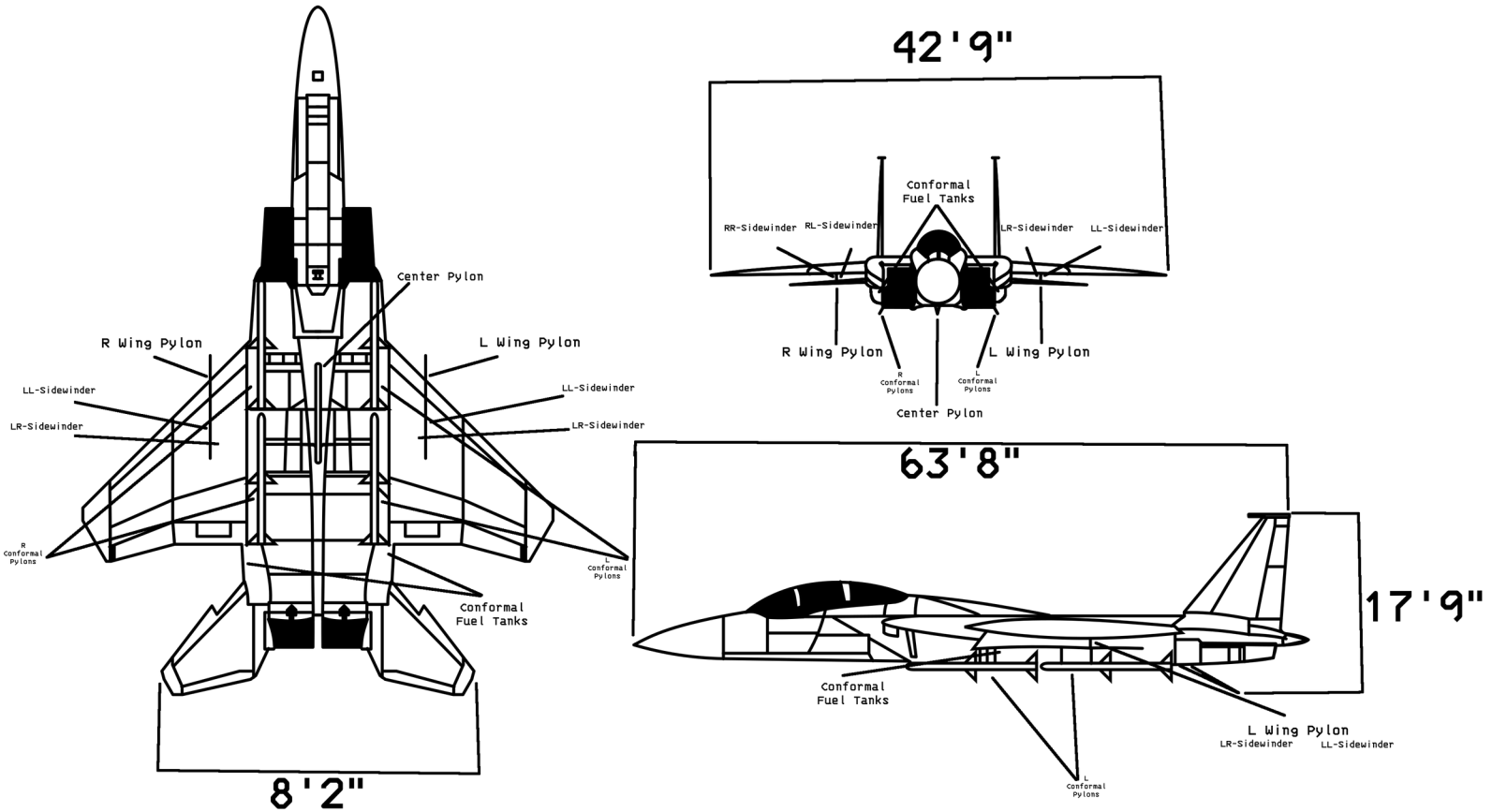
The following audio warnings are given:

- AOA > 25 (900hz beep beep pause)
- AOA > 30 and gear down (1,600hz beeping)
- 85% to 91% of maximum G loading reached (900hz 4 times per second)
- 92% to 95% of maximum G loading (900hz 10 times per second)
- 96% to 100% of maximum G loading (continuous 900hz tone)
- Gear not down when below 200 kts. (900hz beeping)
- Voice warnings for:
 - Whenever the Master Caution light illuminates (CAUTION)
 - Bingo Fuel
 - More than 100% of maximum G loading (OVER-G)

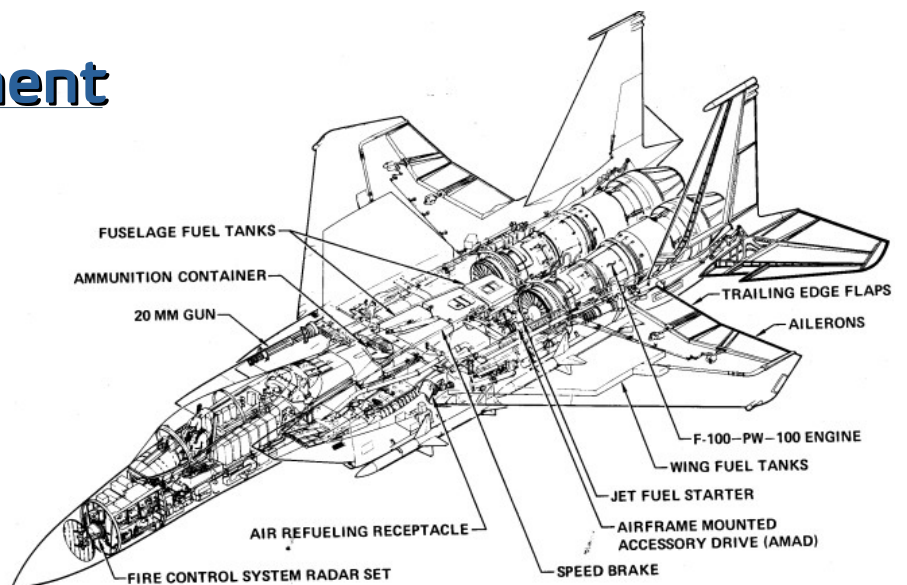
The OWS calculates the maximum permissible G loading based upon the current configuration.

F-15's Specifications

Airframe



General Arrangement



Characteristics Table

Data	F-15C/D	F-15EX
Crew	<i>C</i> : 1 <i>D</i> :2	2
Length (nose-tail)	63ft 8in	63ft 8in
Wingspan	42ft 9in	42ft 9in
Height	17ft 9in	17ft 9in
Wing area	608 ft ²	608 ft ²
Airfoil (root)	NACA 64A006.6	NACA 64A006.6
Airfoil (tip)	NACA 64A203	NACA 64A203
Empty weight	30,000 lbs	30,000 lbs
Gross weight	44,000 lbs	44,000 lbs
Max takeoff weight	≈ 70,000 lbs	≈ 85,000 lbs
Fuel Capacity (no external tanks)	14,043 lbs/2,229 gal - 15,589 lbs/2,289 gal (at true max)	14,043 lbs/2,229 gal - 15,589 lbs/2,289 gal (at true max)
Fuel Capacity (with external tanks)	15,589 lbs/3,993 gal – 27,172 lbs/4,313 gal (at true max)	15,589 lbs/3,993 gal – 27,172 lbs/4,313 gal (at true max)
Fuel Capacity (CFTs) - each	4,586 lbs/728 gal - 4,999 lbs/793 gal (at true max)	4,586 lbs/728 gal - 4,999 lbs/793 gal (at true max)
Powerplant	2x F-100-PW-220	2x F-110-GE-129

Performance Table

Data	F-15C/D	F-15EX
Max speed (at altitude)	1.7 Ma – 1,200 kts	2.6 Ma – 1,750 kts
Max speed (at sea level)	1.1 Ma – 750 kts	1.5 Ma – 1,000 kts
Cruise speed	490 kts	Btw 490-520 kts
Range (three ext fuel tanks)	2,144 nmi	2,144+ nmi
Combat range (counter mission 1 ext fuel tank)	586 nmi	586+ nmi
Combat range (4x AIM-7, internal fuel)	470 nmi	470+ nmi
Combat range (no fly zone, 3 ext fuel tanks)	1,061 nmi	1,061+ nmi
Ferry range (CFTs and 3 ext fuel tanks)	3,000 nmi	3,000+ nmi
Service ceiling	45,000 ft	60,000 ft
G limits	+9	+9
Rate of climb (with pylons)	67,050 ft/min	≈ 75,000+ ft/min
Thrust/weight ratio	1.07 – 1.26 (with loaded weight and 50% internal fuel)	1.07 – 1.26 (with loaded weight and 50% internal fuel)
Radar range	Min: 5nmi, Max: 200nmi	Min: 5nmi, Max: 200nmi

Payload

Data	F-15C/D	F-15E
Guns	1x 20mm M61A1 Vulcan, 940 rounds	1x 20mm M61A1 Vulcan, 940 rounds
Countermeasures	Up to 120 flares and 150 chaffs	Up to 120 flares and 150 chaffs
Hardpoints count	11	25
Pylons count	3 (2 wings, 1 central)	3 (2 each wing, 1 central) + 2 fuselage + 3 each CFT
Missile rails count	4 (2 each wing pylon)	8 (2 each wing pylon)
CFTs	2	2
A/A Missiles	AIM-7F, AIM-9L, AIM-120C	AIM-7F, AIM-9M, AIM-9X, AIM-120D
Bombs	MK-82, MK-83, CBU-87	MK-82, MK-83, MK-84, CBU-87, CBU-105, B61-12
Other	N/A	LAU-68C (A/G unguided Hydra 70 rockets)

Armaments

A/A Missiles

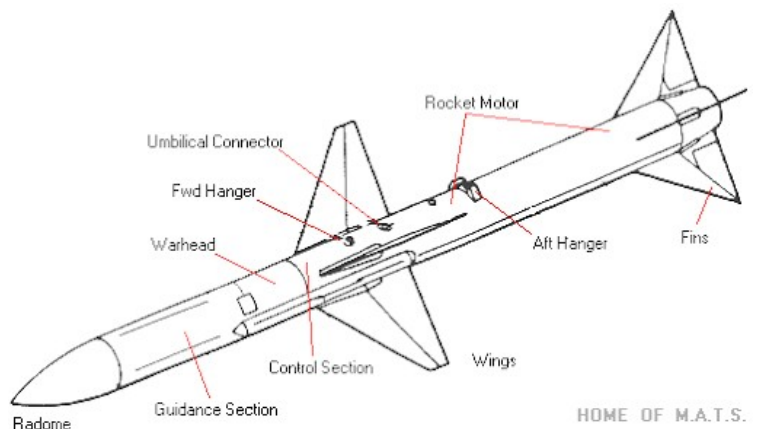
The F-15 can carry most of the USAF's A/A missiles, being AIM-7s, AIM-9s and AIM-120s. The F-15EX is also capable to carry to-come new AIM-260s, though they're not released yet and therefore we don't know the AIM-260's specifications and I didn't coded them in the model.

The F-15 can carry the following variants of the missiles:

- AIM-7M
- AIM-9L
- AIM-9M
- AIM-9X (F-15EX only)
- AIM-120C
- AIM-120D (F-15EX only)

AIM-7 Sparrow

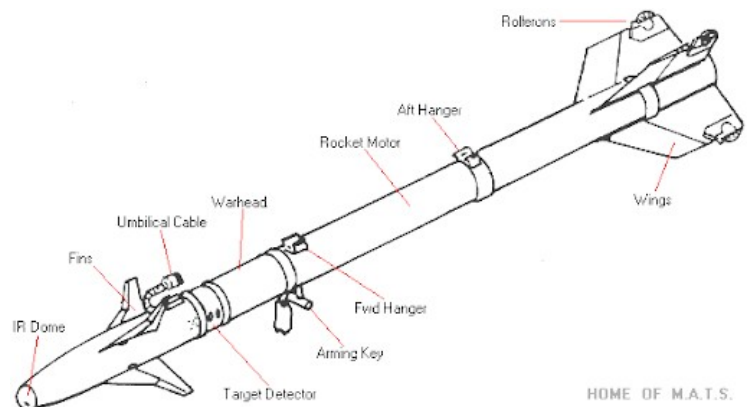
The AIM-7 Sparrow is an USAF medium-range semi-active radar homing air-to-air missile. AIM-7s were the principal beyond visual range (BVR) A/A missiles of the West in the late 1950's, until the 1990's, where it got replaced by the AIM-120 AMRAAM. They are though still in service. The AIM-7F, is the variant currently coded in the F-15 module. It can be fitted in the F-15 in its four body conformal stations. The NATO code for the AIM-7 is "FOX 1".



Semi-active radar homing : means that only the shooting aircraft's radar will be tranceiving the target's informations (altitude, speed, heading etc.). Whereas in an active-radar homing missile, both the shooting aircraft and the missile will.

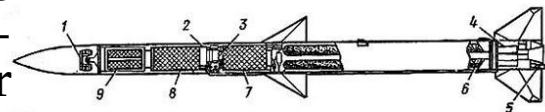
AIM-9 Sidewinder

The AIM-9 Sidewinder is an USAF short-range infrared homing air-to-air missile. The AIM-9 is one of the oldest, cheapest and most successful A/A missiles, its latest variants remaining standard in Western-allied forces. It's the standard missile used for dogfighting in the USAF. It's the AIM-9M and AIM-9X variants of the missile that are coded in the F-15 module. It can be fitted in the left and right wing's pylons' sidewinder stations. The NATO code for the AIM-9 is "FOX 2".



AIM-120 AMRAAM

The AIM-120 AMRAAM is an USAF medium-to-long-range active-radar homing beyond-visual-range (BVR) air-to-air missile. It's now the standard for air-to-air interceptions in the Western-allied forces, replacing both AIM-7s and AIM-54s. Its NATO code is "FOX 3". It's both AIM-120C and newer AIM-120D variants that have been coded for the F-15 module.



- 1 - antenna;
- 2 - inertial block;
- 3 - radar fuse;
- 4 - rudder drives;
- 5 - command line receiver antenna;
- 6 - engine;
- 7 - warhead;
- 8 - block of electronic equipment;
- 9 - transmitter and power supply batteries.

A/A Missiles Stats Comparison

Data	AIM-7F	AIM-9L	AIM-9M	AIM-9X	AIM-120 B	AIM-120D
Weight	510 lbs	196 lbs	196 lbs	196 lbs	291 lbs	335 lbs
Length	12'	9'11"	9'11"	9'11"	12'	12'
Warhead	88 lbs	21 lbs	21 lbs	21 lbs	44 lbs	39 lbs
Guidance	Semi-radar	Heat	Heat	Heat	Active Radar	Active Radar
Operational range	.16 – 22 nmi	.2 – 9.7 nmi	.16 – 12.5 nmi	.16 – 15 nmi	.35 – 38.8 nmi	3 – 123 nmi
Stage 1 thrust	5,750 lbf	2,660 lbf	2,880 lbf	2,880 lbf	2,700 lbf	2,970 lbf
Stage 2 thrust	1,018 lbf	0 lbf	0 lbf	0 lbf	280 lbf	308 lbf
Optimal altitude	30,000 ft	25,000 ft	25,000 ft	25,000 ft	30,000 ft	30,000 ft
Optimal closing	Ma 2	Ma 2	Ma 2	Ma 2	Ma 2	Ma 2
Max G-load	20 G	32 G	45 G	50 G	30 G	40 G
Seeker field degrees	110°	82°	82°	180°	135°	155°
FCS field degrees	25°	55°	67°	180°	110°	50°
Compatibility	All variants	C/D variant only	EX variant only	EX variant only	C/D variant only	EX variant only
Max loadout	4	4	16	16	8	28

A/G Bombs

The F-15 can carry most of the a wide range of the USAF's A/G bombs, being compatible with the MK-bombs family, cluster CBU bombs, GBU guides bombs, and B61 nukes.

The F-15 can carry the following variants of the bombs:

- MK-82
- MK-83
- MK-84
- CBU-87
- CBU-105 (F-15EX only)
- GBU-12 (F-15EX only)
- GBU-24 (F-15EX only)
- GBU-31 (F-15EX only)
- GBU-54 (F-15EX only)
- B61-12 (F-15EX only)

MK-82

The MK-82 is a general-purpose unguided bomb used by the USAF and allied forces. It's part of the Mark 80 series, along the MK-83 and MK-84. The MK-82 is an unguided bomb, though it can be upgraded to become the BGU-12 and allow laser guidance to be operated on it. It weighs 500 lbs, making it the lighter variant of the Mark 80 series. It uses Tritonal (a mix of TNT and aluminum powder) as explosive.

MK-83

The MK-83 is a general-purpose unguided bomb used by the USAF and allied forces. It's part of the Mark 80 series, along the MK-82 and MK-84. The MK-83 is an unguided bomb, though it can be upgraded to become the BGU-24 and allow laser guidance to be operated on it. It weighs 1,000 lbs, making it the medium-weight variant of the Mark 80 series. It uses Tritonal (a mix of TNT and aluminum powder) as explosive.

MK-84

The MK-84 is a general-purpose unguided bomb used by the USAF and allied forces. It's part of the Mark 80 series, along the MK-82 and MK-83. The MK-84 is an unguided bomb, though it can be upgraded to become the BGU-31 and allow laser guidance to be operated on it. It weighs 2,000 lbs, making it the heavier variant of the Mark 80 series. It uses Tritonal (a mix of TNT and aluminum powder) as explosive.

B61-12

The B61-12 is a variable-yield nuclear bomb that incorporates updated technology for more precise targeting, improved safety features, and greater reliability. The simulated variant's blast can go up to 1 mile in diameter.

CBU-87

The CBU-87 is an A/G cluster bomb designed for anti-armor, anti-personnel and incendiary effects. It's commonly used by the USAF and allied forces. It weighs about 950 lbs and clusters 202 BLU-97/B submunitions to an area varying from 218 to 328 yrd.

CBU-105

The CBU-105 is an A/G cluster bomb designed for tank destruction and heavy-armored vehicles. It's commonly used by the USAF and allied forces. It weighs about 930 lbs and clusters 10 BLU-108 submunitions to an area varying from 109 yrd. The CBU-105 is a more advanced version of the CBU-87, using smart skeet warheads in its submunitions, each able to detect and track moving vehicles.

A/G Bombs Stats Comparison

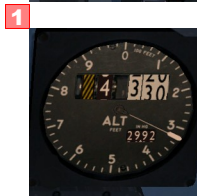
Data	MK-82	MK-83	MK-84	CBU-87	CBU-105	B61-12
Weight	500 lbs	1,014 lbs	2,039 lbs	950 lbs	934 lbs	775 lbs
Length	7'6	9'11"	12'9"	7'7"	7'8"	11'8"
Warhead	192 lbs	445 lbs	945 lbs	128 lbs (<i>cluster submunitions not simulated</i>)	136 lbs (<i>cluster submunitions not simulated</i>)	825 lbs
Guidance	Unguided	Unguided	Unguided	Unguided	GPS (<i>can be unguided</i>)	GPS (<i>can be unguided</i>)
Navigation	direct	direct	direct	direct	APN	APN
Arming Time	4 sec	5 sec	6 sec	3 sec	3 sec	30 sec
Drag coeff	.125	.125	.125	.14	.14	.125
Max G-load	1.6 G	1.8 G	2.0 G	1.8 G	1.8 G	2.0 G
Max pitch degrees	-90°	-90°	-90°	-90°	-45°	-45°
Max blast yield (diameter)	164 yrd	191 yrd	246 yrd	213 yrd	108 yrd	1,859 yrd (1 mi or .8 nmi)
Ejector speed	20'/sec	10'/sec	5'/sec	10'/sec	10'/sec	15'/sec
Racks	X3 – X5	X1 – X3	X1 – X3	X3	X 1 – X3	X1
Compatibility	All variants	All variants	All variants	All variants	EX variant only	EX variant only
Max loadout	21	21	8	21	13	1

F-15 C/D

COCKPIT INSTRUMENTS



MAIN CENTER PANEL



Altimeter

The Altimeter shows the barometric pressure altitude and is displayed in units of 20 feet.



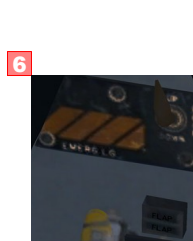
Fire Suppression Panel

There is no simulation of the fire suppression system. The burn thru lights indicate a flameout in the engines. The three red buttons below it are used to suppress engine flameout.



Acceleration (G)

The Accelerometer shows the current values of positive and negative G loads. The display of maximum +G and -G is not simulated.



Flaps Panel

The flaps panel contains the hook control for the emergency hook. This is intended for use in wire arrested emergency airfield landings and is not strong enough for carrier landings; however in this model these limits are not simulated so it is possible to land the aircraft on a carrier. There is an indicator for the status of the flaps. The top light will indicate in amber when the flaps are in transit, and the green light below is when the flaps are locked in the open position.



Angle of Attack (AOA)

The AoA indicator is positioned on the instrument panel under the IAS and Mach meter. It is used for indicating the current AOA value within the limits of 0 to 45 units. The AOA indicated values do not correspond to actual degrees. In the area of the landing AOA (20-22 units) there is a corresponding index on the indicator.



Airspeed Indicator (IAS)

The IAS and Mach meter is positioned to the right of MPCD. It shows the indicated airspeed and the Mach number. The fixed scale of the indicated airspeed is graduated within the limits of 50 to 1,000 kts. The moving scale of the Mach number shows the value of Mach number within the limits of the operating altitudes and speeds. Mach numbers are shown starting from the indicated speed value of 200 kts.



Landing Gear Panel

The landing gear panel contains a handle that controls the position of the landing gear. Landing gear can be deployed or retracted by clicking on the handle (or by using g/G). When the gear is in transit a red lamp will illuminate the handle. When on the ground the handle will move but the gear will not. There is a landing gear warning horn that will sound when below 200kts and descending; this can be silenced by pushing the silence button.

Next to the landing gear is the pitch ratio panel. The primary flight control effectivity is adjusted following a schedule to produce about the same acceleration (G) per inch of stick movement. The gauge indicates the current ratio, and the emergency pitch ratio switch will lock this at 0.5 in the event of system failure. When the landing gear is down the pitch ratio is locked at 1.0.

8

Emerg Jett Panel



The Emerg Jett button will, when fully depressed, result in the jettison of pylons, whether it's armament or fuel tanks.

The Emergency Jettison panel also contains a selector for the Steer Mode which is primarily used to control the flight director bars on the ADI.

13



Red Warnings Panel

This panel contains several warning indicators:

- "Canopy Unlocked": on when the canopy is fully open
- "Program": not simulated
- "Minimum": on when either flare or chaff level is below 36 to indicate the level is low
- "Chaff": on when chaffs are being released
- "Flares": on when flares are being released

14



ARM Modes Panel

This selects the primary arm mode. For normal air-to-air operations, all modes require to be offline. For air-to-ground operations, A/G mode needs to be online. Other modes are not simulated.

9



Lower Center Panel

On the top center, you have the Attitude Direction Indicator, providing a 3D flight trajectory cue to maintain the desired path. The cue takes form of V steering bars. Below the that, you have the Horizontal Situation Indicator, combining a heading indicator with a VHF omnidirectional range-instrument landing system display. With the "heading set" switch you can set the heading and with the "course set" switch you can set the course. You also find the emergency jett panel on the right and the master arm modes on the left.

10



Standby Instruments

There are standby instruments for the following:

- ADI
- Airspeed
- Altitude

These will operate when no electrical power is available or in the event of Avionics failure. The lighting for these is controlled by the switch "STBY INST" on the lighting panel.

11



Vertical Velocity Indicator (VVI)

The VVI is used for indicating the vertical aircraft speed, i.e. the climb and sink rate in thousands feet per minute. When the indicator arrow moves in a clockwise direction, it indicates that the aircraft is increasing its flight altitude. When the indicator arrow moves in a counter-clockwise direction, it indicates that the aircraft is descending.

12



Cabin Pressure and JFS

Providing that the ECS is receiving sufficient bleed air the cabin pressure will be maintained according to the correct schedule.

To start JFS first use the Engines Master Panel in the right hand console to switch the JFS on and then pull the lever. Providing that there is sufficient hydraulic pressure in the Util system JFS will start. If there is insufficient pressure you must use the Quickstart button from the menu as there is no way to recharge this system without engines or external electrical power.

15



Caution Panel

This panel contains warning lights that indicate problems with the aircraft. The Emergency Vent lever will rapidly depressurize the cabin in case of emergency.

Each different warning will not be listed and explained in this manual.

Main panel / Engines



The right hand side of the main panel contains the engines gauges, together with the hydraulics and fuel gauges.

The drums on the EGT gauges will have the red digits for the first two digits when EGT is dangerously high.

The UTIL hyd system needs to be over 1,200 psi to perform a JFS start.

The pressures displayed will normally be 2,900–3,100, however these values may differ when an engine has flamed out, or when the emergency (electrical) hydraulic backup is running in the case of primary pump failure. When the hydraulic pressure drops below 1,000 the primary surfaces, gear and flaps will not move. This only usually happens when fuel runs out and N1 drops below 20%.

The selector on the fuel gauge selects the tanks displayed in the lower two drum displays. This selector also has a test position that is spring loaded. To ensure that the gauge is operating correctly select this position and the displays will show 6,000.

Left Console

The throttles move roughly half way for military power, and the remaining area is for the five stage thrust augmentation (Burners).

The lower switch on the throttle controls the selected armament type, and cycles between GUN, SRM (Short Range Missiles AIM-9), MRM (Medium Range Missiles AIM-7, AIM-120)

1

This is the external lighting panel. It contains intensity for formation lights, positions lights and anti-collision lights.

2

This is the RADIO 2 configuration panel. You can use the circular switches to change the frequency and channel.

3

This is the autopilot/axis control panel. It allows you to hold yaw, roll and pitch.

4

This is the fuel panel. It allows you to enable fuel dump.

5

This is the radar configuration panel. With the power switch, you can either put the radar offline, set it to standby (RWS) or operational (TWS).

The switches on the bottom part of the panel are switches for the radar too. For the radar to work properly, you usually set the MASTER switch to normal, the M3/A switch to ON, and both MODE 4 switches to ON.



Right Console

This right console contains 5 different panels, which are not all developed. It contains, oxygen and cabin pressure configuration, master engine switches, internal lighting, TEWS radar configuration and autopilot information.

1

This is the master engine switches. The STARTER switch sets the Jet Fuel Starter (JFS) ready, indicated by a green light. The GEN and EEC, both left and right engine switches are turned on once both engines are online and ready. The engine master switches are secured by a cache. They're set on one by one once JSF is online.

2

This is the oxygen regulator panel. The SUPPLY switch is used as a cutoff switch for oxygen's flow. When online, the oxygen system provides oxygen. It's set online during normal flight, and offline when on ground or when switching to a backup system. The oxygen supply pressure gauge should indicated a value inside the bold area. On the right, on the ECS panel, you have an OXY TEST button, used to perform an oxygen flow test. The gauge next to it indicates the amount of liquid oxygen in metric liters.

3

This is the cabin temperature panel. The only simulated feature is the windshield. It's used to remove ice on the canopy glass, reducing vision, usually occurring at high altitudes or when external temperature is about below 32 degrees.

4

This is the internal lighting panel. Each switch controls a panel or console's lighting intensity.

5

This is an information panel for the autopilot. This isn't currently simulated.

6

This is a config panel for the TWS radar mode. This isn't currently simulated.



HUD Controls

At the top is the IFF / Transponder code. Underneath is the radio 1 comms frequency and mode. At the bottom there are the brightness and symbology controls.

F-15 DISPLAYS



The HUD (Heads-Up-Display)

The F-15 HUD is a multi-function device that will display different information depending on the primary mode as determined by the avionics. Prior to the E variant the HUD used an optical combiner with two panes of glass angled such that at the correct eyepoint these appear as a single device. This is accurately simulated, so when viewing off angle the two panes will be visible. All moded variants have the same HUD display.

When Master ARM is on the currently selected weapon type is displayed with the currently selected targetting information on the right, and the weapon identification. Targets are displayed on the HUD as a box. The currently selected target is a double box.

The current vertical G loading is displayed on the bottom left, with the current maximum permitted G loading displayed next to this. If the Overload Warning System is inoperative "NOWS" will be displayed in place of the maximum loading.

The route manager is integrated into the autopilot and when a route is being followed the route manager the routing information will be displayed on the right.

